

Scalability & Future Plan

Date	28 June 2024
Team ID	
Project Name	Smart lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to a single ML model	Reduces flexibility in predictions	High
2	Local model and dataset storage	Difficult to scale for large datasets	Medium
3	Basic security implementation	Needs stronger authentication and data protection	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy on cloud with load balancing to support more users
2	Data Storage	Dataset stored locally in CSV format	Use a cloud database such as MySQL or PostgreSQL
3	Performance	Single-server Flask deployment	Optimize model and deploy with Gunicorn/Nginx for faster response
4	Security	Basic input validation	Implement user authentication, HTTPS, and secure API access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase1	Deploy the application on IBM Cloud	Next 1 Month	Public accessibility and improved availability
Phase2	Add user authentication and loan history	Next 2 Month	Improved security and user management
Phase3	Integrate real-time banking and credit score APIs	Next 3 Month	More accurate and reliable loan predictions
Phase4	Develop a mobile application and multilingual support	Next 4 Month	Better accessibility and wider user adoption